

Counterparty Credit Risk Summary

Valuation date 2026-09-18. Generated by `counterparty-credit-risk/07_report.R`. Data: FRED. 5,000 paths, monthly grid.

Key observations

- **Book:** 7 trades with 4 counterparties, net MTM \$1.91mm. Largest peak PFE (95%) is Airline C at \$9.01mm, reached after 0.7 years.
- **Collateral:** for Global Bank A and Energy Hedge Fund D, the CSA changes peak PFE from \$3.09mm / \$8.87mm (no CSA) to \$953.3k / \$6.76mm. Thresholds and MTAs limit how much it helps.
- **CVA:** total \$165.3k. The largest charge is Airline C (\$80.5k), driven by BB credit and uncollateralised exposure.
- **Wrong-way risk:** linking credit to oil moves Fund D's CVA by 60% (wrong-way) and Airline C's by -42% (right-way). Same trade direction, opposite credit link, opposite result.
- **EAD:** SA-CCR totals \$35.49mm versus \$9.55mm from the simulation (1.4 x EEPE). SA-CCR is more conservative here.
- **Capital:** IRB default-risk RWA \$23.78mm plus BA-CVA RWA \$25.29mm = \$49.07mm total counterparty RWA.
- **Backtest:** rates 21d: tails too thin (underestimates risk); rates 63d: tails too thin (underestimates risk). The usual fix is fatter-tailed shocks or regime-dependent volatility.
- **Validation:** all at-market trades reprice to ~0; 8 of 8 martingale checks pass; CVA Monte Carlo error is within 2% of the estimate for every counterparty.

1. Trade book

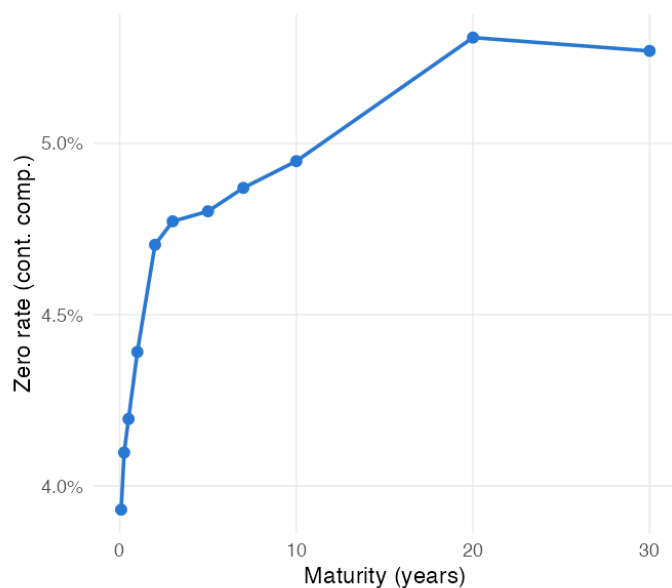
Trade	Counterparty	Description	Strike	MTM	DV01
T1	Global Bank A	5y payer swap (bank pays fixed)	4.601%	\$550.0k	\$22.3k
T2	Global Bank A	10y receiver swap (bank receives fixed)	5.085%	\$234.9k	-\$24.1k
T3	Global Bank A	1y EUR forward, bank buys EUR	1.1800	-\$223.6k	\$2.3k
T4	Industrial Corp B	7y swap: corporate hedges floating debt	5.315%	\$940.4k	-\$24.3k
T5	Industrial Corp B	2y EUR forward: importer buys EUR from bank	1.1861	-\$163.6k	-\$3.2k
T6	Airline C	2y jet-fuel proxy hedge: airline pays fixed	83.5701	\$2.28mm	-\$0.5k
T7	Energy Hedge Fund D	18m swap: fund is long oil, bank short	79.3068	-\$1.71mm	-\$0.1k

2. Market data and model calibration

Parameter	Value	Source
Hull-White mean reversion a	0.05	assumption (config)
Hull-White sigma (abs. rate vol)	0.006506	1y UST, last 3y
EURUSD volatility	0.06725	last 3y
WTI mean reversion κ	1.574	AR(1), last 10y
WTI half-life (years)	0.4403	$\log(2)/\kappa$
WTI long-run level (\$/bbl)	67.97	$\exp(\alpha)$
WTI volatility	0.525	AR(1), last 10y
EUR rate (ECB deposit)	0.025	FRED ECBDFR
EURUSD spot	1.146	FRED DEXUSEU
WTI spot (\$/bbl)	101.4	FRED DCOILWTICO

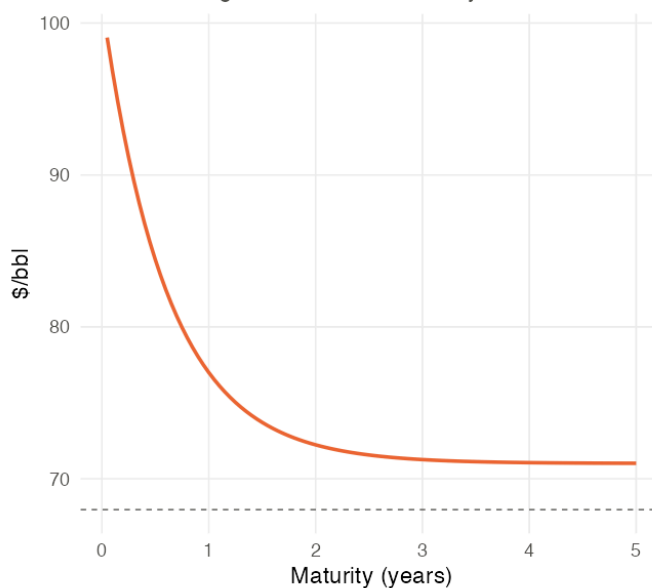
US Treasury zero curve

As of 2026-09-18

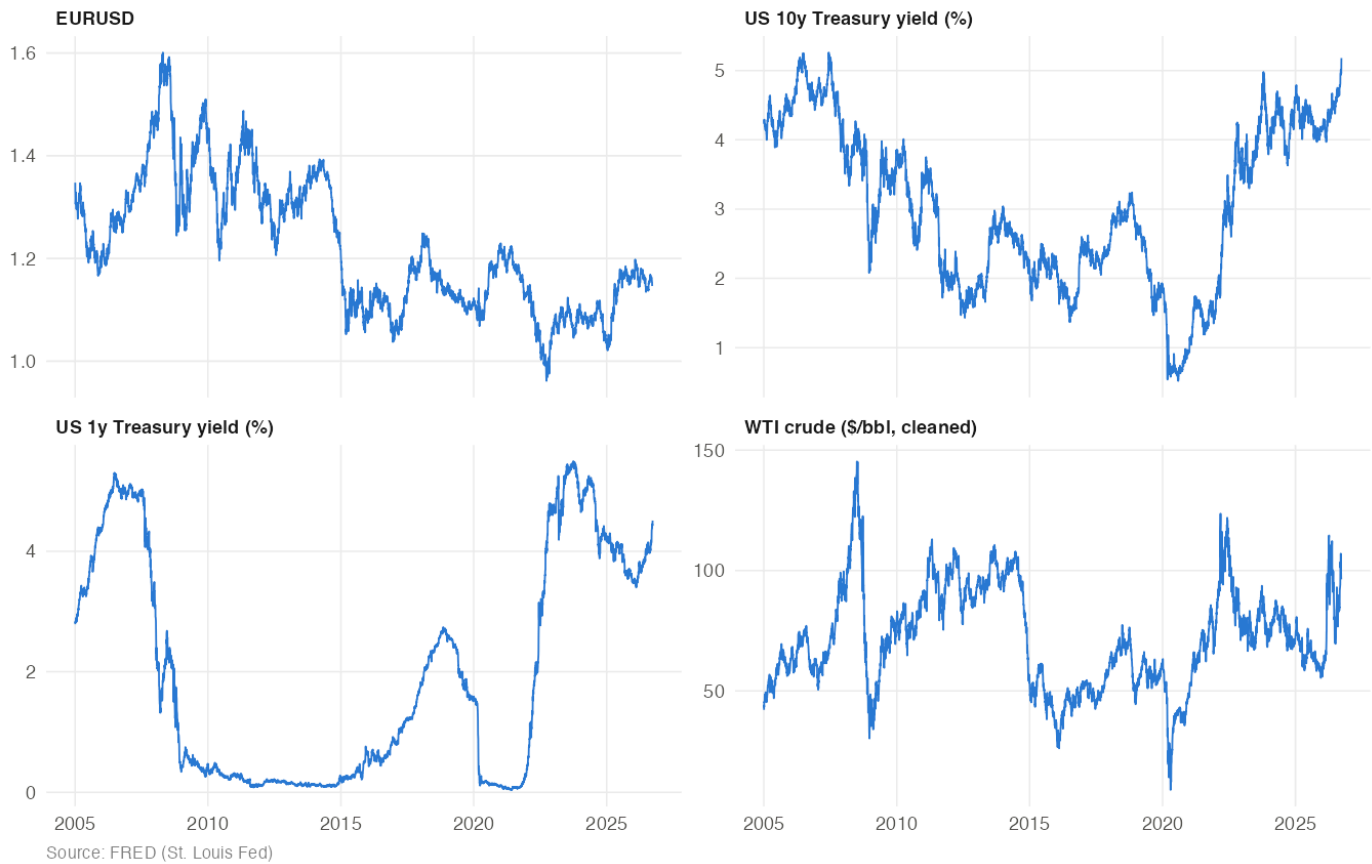


WTI model forward curve

Dashed = long-run level. Half-life 0.4 years.



Risk-factor history used for calibration and backtesting

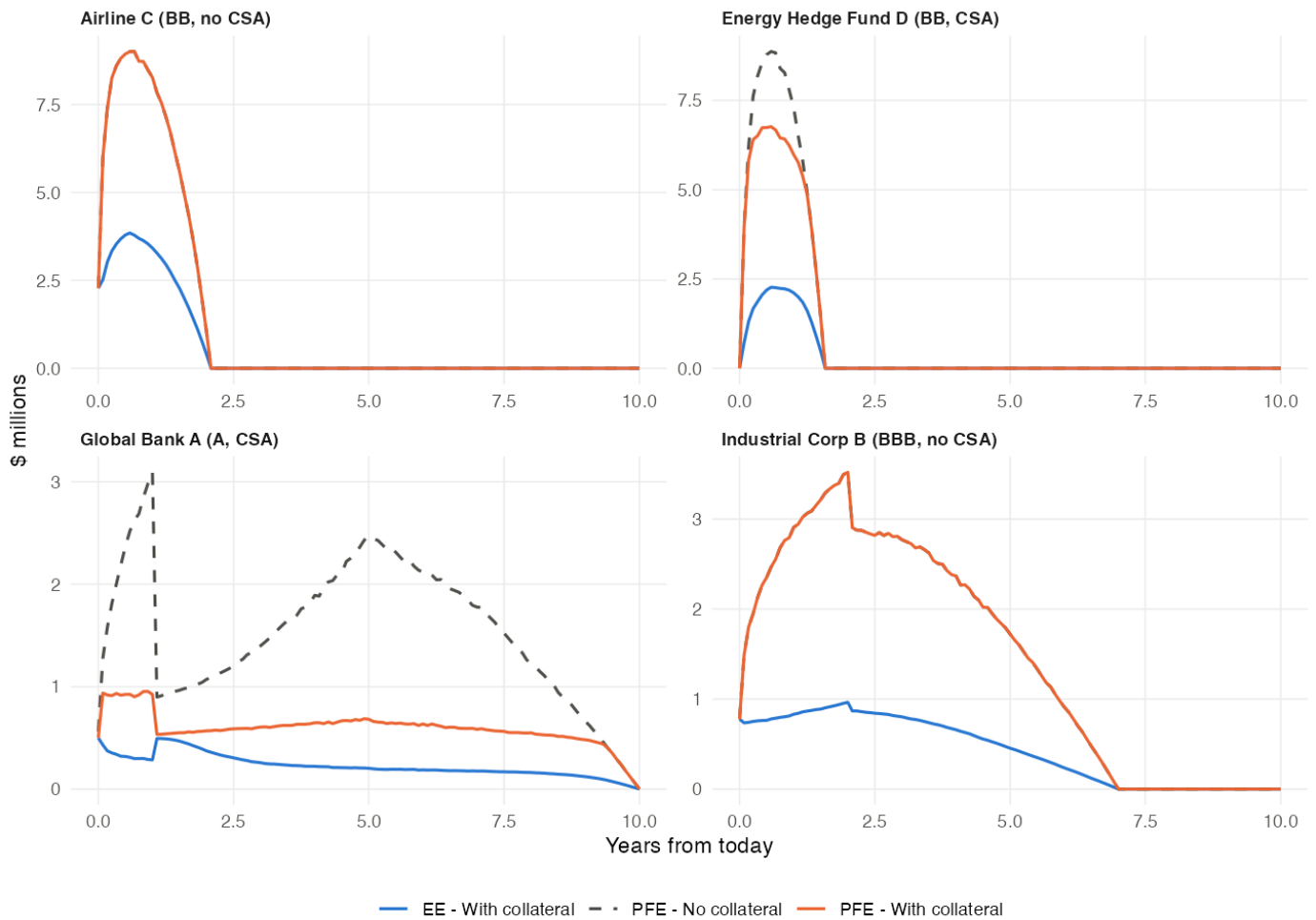


3. Exposure

Counterparty	Rating	CSA	MTM	Peak EE	Peak PFE	Peak PFE (no CSA)	EEPE
Global Bank A	A	Yes	\$561.3k	\$500.0k	\$953.3k	\$3.09mm	\$500.0k
Industrial Corp B	BBB	No	\$776.9k	\$964.5k	\$3.52mm	\$3.52mm	\$789.3k
Airline C	BB	No	\$2.28mm	\$3.85mm	\$9.01mm	\$9.01mm	\$3.58mm
Energy Hedge Fund D	BB	Yes	-\$1.71mm	\$2.27mm	\$6.76mm	\$8.87mm	\$1.95mm

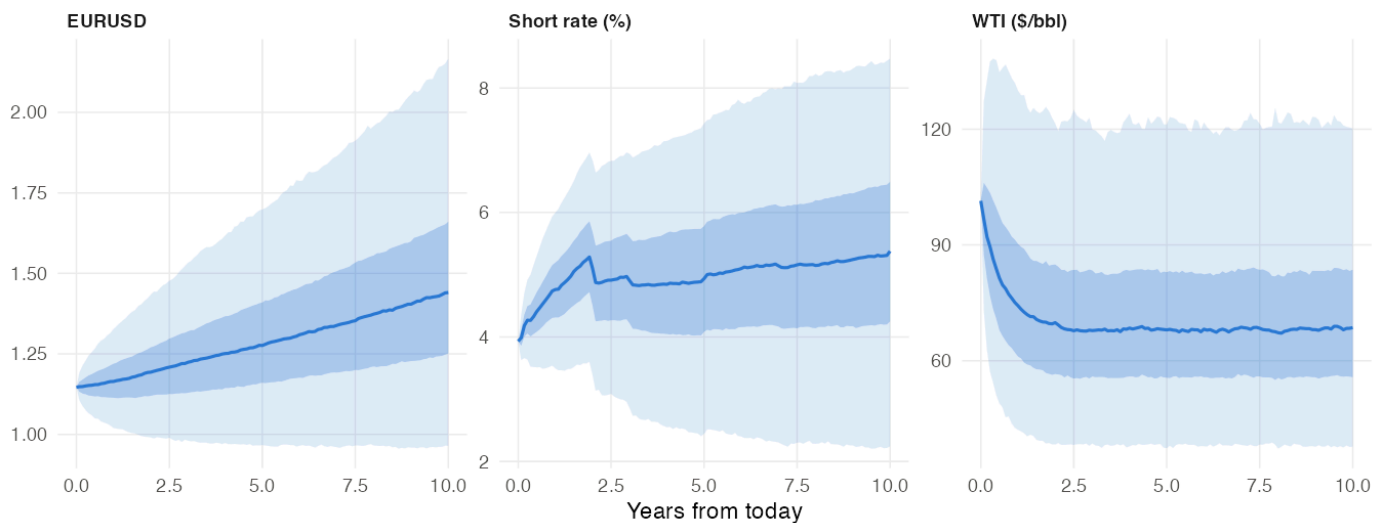
Exposure profiles: expected exposure (EE) and 95% PFE

Swaps: 'hump' shape (uncertainty grows, then cash flows run off). CSA caps exposure (dashed = without it).



Simulated risk factors: median, 50% and 95% bands

Oil mean-reverts (half-life 0.4 years), so its band levels off; FX bands keep widening with $\sqrt{\text{time}}$.

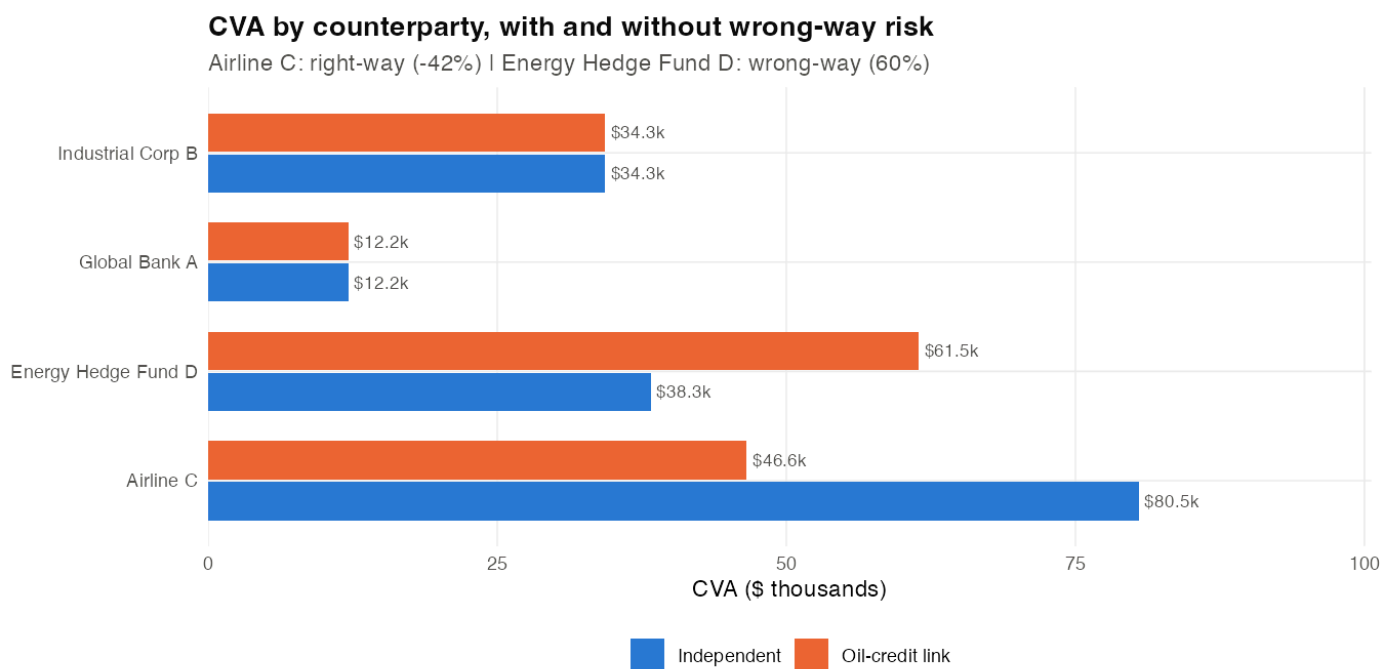


Stressed current exposure (MTM after instant shocks):

Counterparty	Base	Rates +100bp	Rates -100bp	EURUSD -10%	Oil -30%	Oil +30%	Combined: rates -100bp, oil -30%, EUR -10%
Global Bank A	\$561.3k	\$667.7k	\$565.9k	-\$1.67mm	\$561.3k	\$561.3k	-\$1.67mm
Industrial Corp B	\$776.9k	-\$1.90mm	\$3.62mm	\$2.41mm	\$776.9k	\$776.9k	\$5.26mm
Airline C	\$2.28mm	\$2.23mm	\$2.33mm	\$2.28mm	\$6.93mm	-\$1.73mm	\$7.00mm
Energy Hedge Fund D	-\$1.71mm	-\$1.71mm	-\$1.70mm	-\$1.71mm	\$5.39mm	-\$7.86mm	\$5.43mm

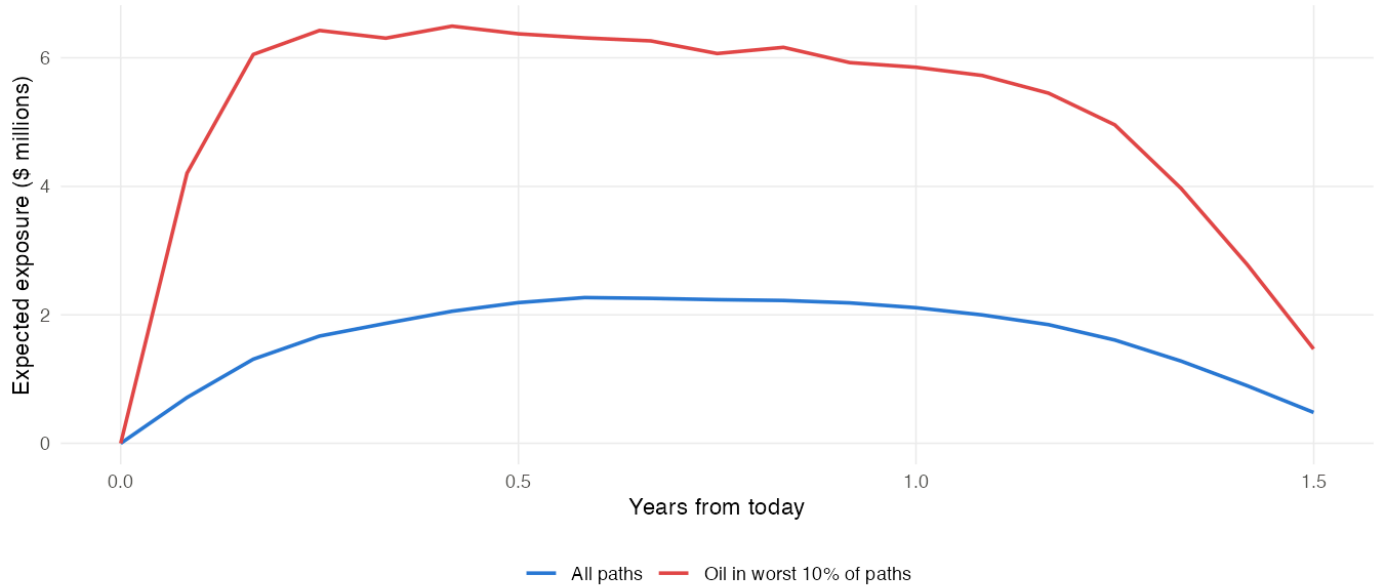
4. CVA and wrong-way risk

Counterparty	Spread (bp)	CVA	CS01	CVA with oil link	WWR
Global Bank A	65	\$12.2k	\$0.2k	\$12.2k	No link modelled
Industrial Corp B	94	\$34.3k	\$0.4k	\$34.3k	No link modelled
Airline C	155	\$80.5k	\$0.5k	\$46.6k	Right-way
Energy Hedge Fund D	155	\$38.3k	\$0.2k	\$61.5k	Wrong-way



Wrong-way risk: exposure to Energy Hedge Fund D when oil falls

At 1 year, expected exposure on low-oil paths is 2.8x the average: highest when an oil-long fund is weakest.



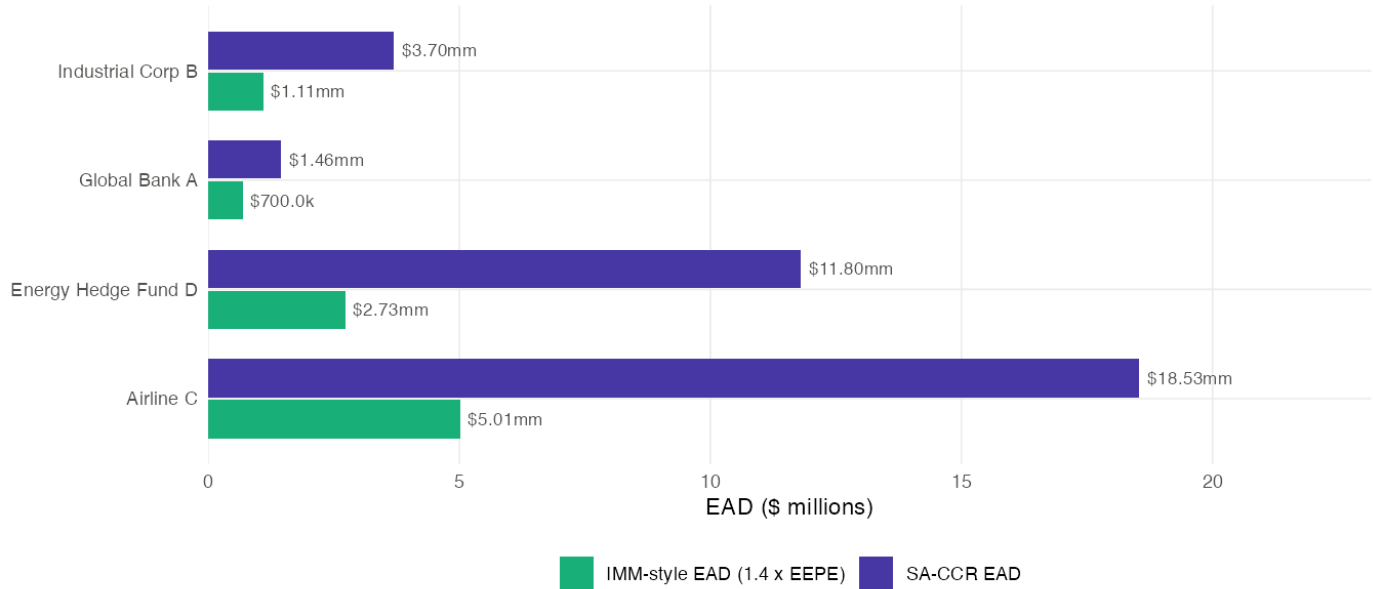
5. Regulatory exposure and capital (illustrative)

Counterparty	RC	PFE add-on	Multiplier	SA-CCR EAD	IMM-style EAD	IRB PD	IRB RWA
Global Bank A	\$500.0k	\$541.6k	1.00	\$1.46mm	\$700.0k	0.05%	\$667.9k
Industrial Corp B	\$776.9k	\$1.87mm	1.00	\$3.70mm	\$1.11mm	0.15%	\$1.93mm
Airline C	\$2.28mm	\$10.96mm	1.00	\$18.53mm	\$5.01mm	0.60%	\$11.44mm
Energy Hedge Fund D	\$5.25mm	\$3.18mm	0.81	\$11.80mm	\$2.73mm	0.60%	\$9.74mm

Measure	Amount
Total CVA	\$165.3k
Total SA-CCR EAD	\$35.49mm
Total IMM-style EAD	\$9.55mm
IRB default-risk RWA	\$23.78mm
BA-CVA capital	\$2.02mm
BA-CVA RWA	\$25.29mm
Total CCR RWA (IRB + BA-CVA)	\$49.07mm

Exposure at default: standardised (SA-CCR) vs. internal model

SA-CCR uses fixed supervisory factors; the simulation-based EAD reflects this portfolio's actual dynamics.



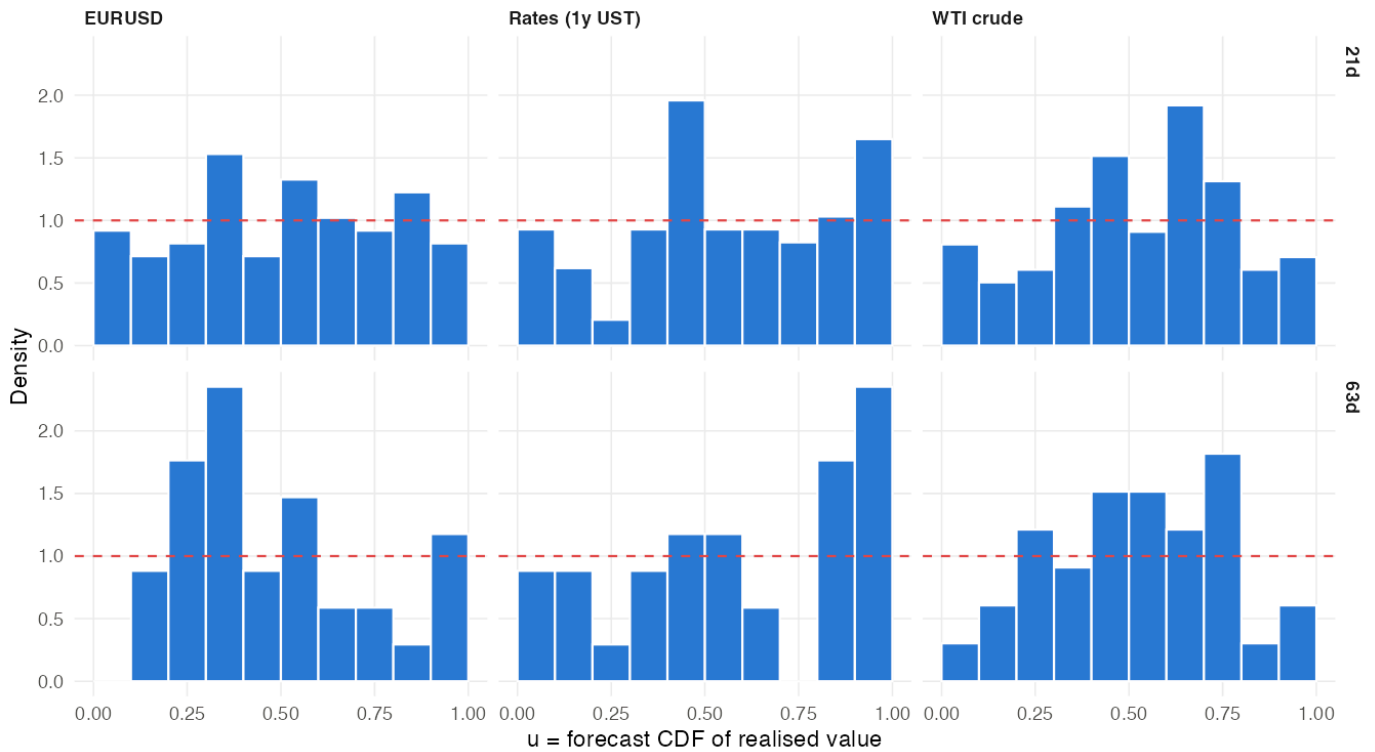
6. Backtesting and validation

Risk-factor PIT backtest (non-overlapping forecasts, rolling out-of-sample calibration):

Factor	Horizon	N	Outside 95%	Expected	Coverage p	KS p	Verdict
fx	21d	98	4	4.9	1.00	0.52	Passes
fx	63d	34	2	1.7	0.69	0.26	Passes
oil	21d	99	4	5.0	1.00	0.12	Passes
oil	63d	33	2	1.7	0.68	0.48	Passes
rates	21d	97	14	4.9	0.00	0.04	Tails too thin (underestimates risk)
rates	63d	34	6	1.7	0.01	0.06	Tails too thin (underestimates risk)

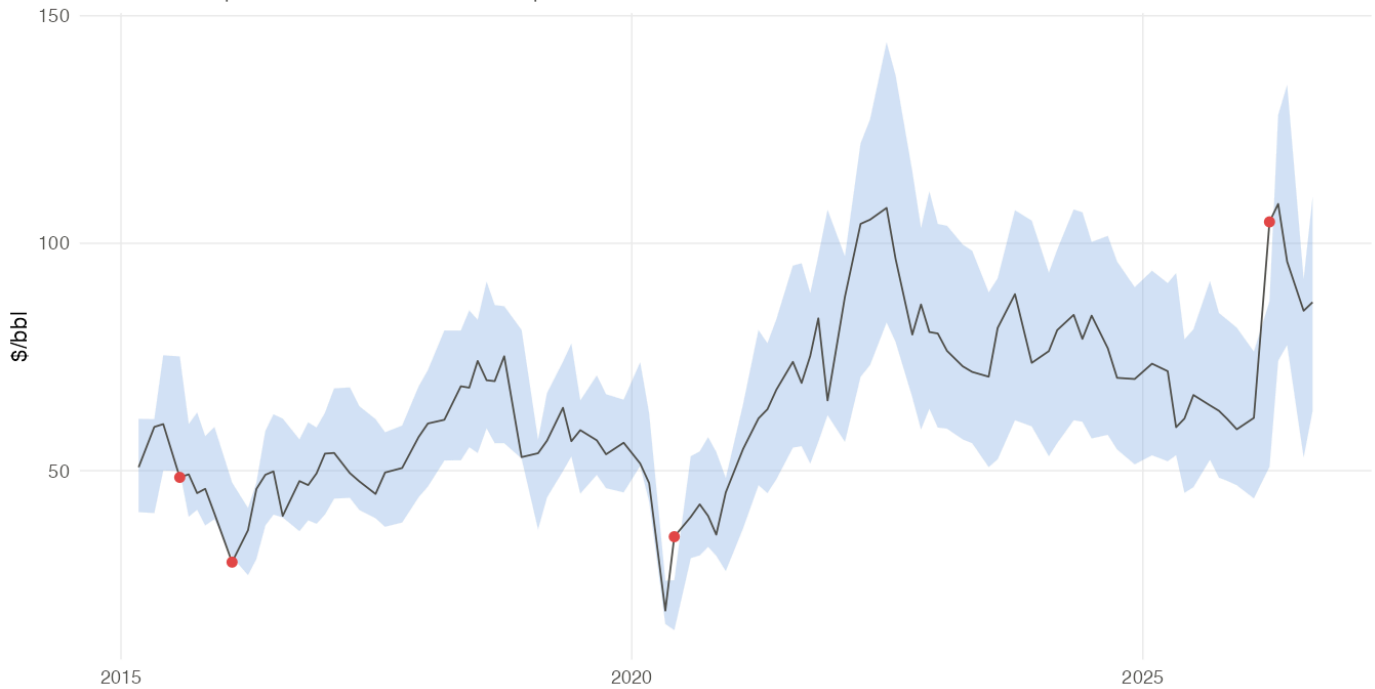
PIT histograms: where did reality land in each forecast?

A good model gives flat bars at the dashed line. Tall end bars = fat tails the model missed.



Oil model backtest: 1-month-ahead 95% forecast band vs. realised WTI

Red: realised price fell outside the band. Expect ~5% of months.

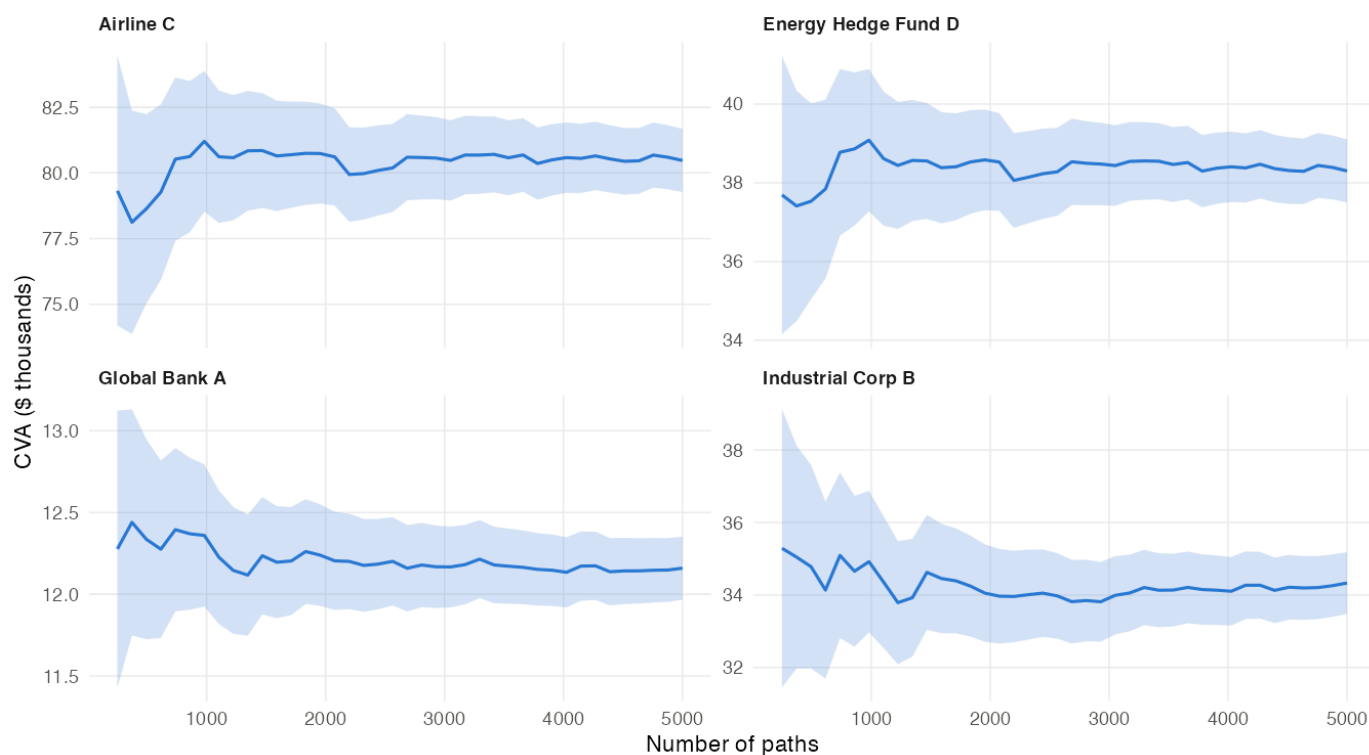


Martingale tests (simulation must reproduce today's prices):

Test	t (yrs)	Model	Market	Error (bp)	Pass
Zero-coupon bond	1	0.96	0.96	-0.79	Yes
Discounted EURUSD	1	1.12	1.12	-12.47	Yes
Zero-coupon bond	2	0.91	0.91	-1.17	Yes
Discounted EURUSD	2	1.09	1.09	-4.60	Yes
Zero-coupon bond	5	0.79	0.79	2.39	Yes
Discounted EURUSD	5	1.02	1.01	44.66	Yes
Zero-coupon bond	10	0.61	0.61	8.42	Yes
Discounted EURUSD	10	0.89	0.89	19.24	Yes

Monte Carlo convergence of CVA

Band = 95% confidence interval. It narrows like $1/\sqrt{\text{paths}}$.



Regulatory notes

- **SA-CCR** (Basel CRE52) is the standardised method for derivative EAD: $\text{EAD} = 1.4 \times (\text{RC} + \text{multiplier} \times \text{AddOn})$. It replaced the older Current Exposure Method; US advanced-approaches banks must use it and other US banks may elect it.
- **IMM** (internal model method) lets approved banks use simulated EEPE instead: $\text{EAD} = \alpha \times \text{EEPE}$, with $\alpha = 1.4$ (or bank-estimated, floored at 1.2). IMM approval requires ongoing backtesting like section 6.

- **CVA capital** (Basel MAR50): banks use BA-CVA (shown here) or SA-CVA (sensitivity-based, built on FRTB machinery). The old internal-model CVA approach was removed.
- **IRB** (Basel CRE31) turns EAD into default-risk capital with the Vasicek one-factor formula. Basel III final rules restrict large corporates and financials to Foundation IRB with supervisory LGDs.
- **Accounting:** CVA (and DVA) are fair-value adjustments under IFRS 13 / ASC 820, so they hit P&L. xVA desks hedge CVA with CDS and market hedges.
- **Timing:** EU applies the final Basel package via CRR3 (FRTB/CVA pieces phased to 2027); UK Basel 3.1 applies from 1 Jan 2027; US agencies re-proposed Basel III endgame in March 2026. Check current status before quoting dates.

Limitations

- One curve (Treasuries) for discounting and forwarding; production uses SOFR OIS curves and a separate EUR curve.
- Hull-White mean reversion is fixed; production calibrates it to swaption volatilities.
- Oil is modelled on spot with a one-factor model; production fits the full NYMEX futures curve (two-factor Schwartz-Smith or similar).
- Collateral uses a simplified VM model (no initial margin, no disputes, threshold + MTA combined).
- The WWR link is a stylised hazard-rate model with an assumed strength (`WWR_B` in config).
- Bond-index spreads stand in for CDS; they include liquidity premia.
- The oil forward curve is the model's own (historical long-run level), not NYMEX futures, so the at-market pricing check proves internal consistency only.
- AR(1) estimates of mean reversion are biased upward in finite samples, so oil's long-horizon volatility may be understated.
- Folding the MTA into the threshold leaves a permanent residual exposure of about the MTA on margined sets (conservative).
- The PIT backtest uses zero drift, so it tests the volatility assumptions rather than the risk-neutral drifts used for CVA.